

Assignment 1: Movie Review Sentiment Classification using Embeddings

Author: Tanjila Ahmed Medha

Date: May 10, 2026

Problem Definition

The goal of this project is to classify movie reviews as positive or negative using text embeddings. Sentiment analysis is a fundamental challenge in natural language processing with applications in recommendation systems, content moderation, and market research.

Dataset

The base dataset is the IMDB movie review dataset, which was processed and filtered to create a custom dataset. Reviews were filtered to a maximum length of 1000 characters to focus on concise, punchy reviews, and a `text_length` feature column was added. The final dataset contains balanced positive and negative samples across train and test splits.

- **Hugging Face Dataset:** <https://huggingface.co/datasets/medhuuu/movie-sentiment-dataset>

Approach

Text embeddings were generated using the `all-MiniLM-L6-v2` sentence transformer model. This model converts each review into a 384-dimensional vector where semantically similar texts are placed close together in vector space. These embedding vectors were then used as input features for three classifiers.

Embedding type: Sentence embeddings (dense, semantic)

Embedding model: `all-MiniLM-L6-v2` (SentenceTransformers)

Results

Classifier	Accuracy	Precision	Recall	F1-Score
Logistic Regression	84.24%	0.84	0.84	0.84
Linear SVM	84.54%	0.85	0.85	0.85
Random Forest	78.99%	0.79	0.79	0.79

The **Linear SVM** achieved the best performance at **84.5% accuracy**, outperforming Logistic Regression slightly and Random Forest by a significant margin. Linear SVM works well with high-dimensional embedding vectors as it finds an optimal hyperplane to separate the classes.

Links

- **GitHub Repo:** <https://github.com/medhuuuu/ir-assignment1-embeddings-system>
- **Hugging Face Dataset:** <https://huggingface.co/datasets/medhuuu/movie-sentiment-dataset>
- **Demo:** <https://huggingface.co/spaces/medhuuu/movie-sentiment-demo>

Reflection on Working with AI

AI tools (Claude) were used extensively throughout this assignment for debugging environment issues, generating boilerplate code, and explaining concepts. The most valuable aspect was getting instant explanations of what embeddings actually do — converting text into vectors where similar meaning means similar position in space. However, AI tools also required careful verification; several suggested commands failed due to my Windows/PowerShell environment and required manual troubleshooting. Understanding each step before running it was essential, as the AI occasionally gave commands that needed adapting to my specific setup.